

Additional file 1: TRIPOD Reporting Checklist

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Completed TRIPOD checklist (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) for the manuscript “*A Multiple Correspondence Analysis Approach to Mitigate the Non-Communicable Disease (NCD) Epidemic: Introducing a Health-Related Behavior-Based NCD Vulnerability Index.*” Section names and appendix letters refer to the main manuscript.

TRIPOD item	Topic	Location in manuscript
1 to 2	Title and abstract	Title page; Abstract
3a, 3b	Background and objectives	Introduction
4a, 4b	Source of data	Data and Variables (Source of Data); Appendix A
5a to 5c	Participants	Data and Variables (Source of Data); Appendix D
6a, 6b	Outcome	Data and Variables (hypertension definition)
7a, 7b	Predictors	Data and Variables (ten NCDVI inputs; confounders)
8	Sample size	Data and Variables (N = 12,650); Appendix D
9	Missing data	Methodology; Source of Data (participant flow)
10a to 10c	Statistical analysis methods	Methodology (MCA; multilevel model)
10d	Validation methods	Methodology (Validation and Sensitivity Analyses)
11	Risk groups	NCDVI tier construction; Results
12	Development vs validation	Methodology; External Validation on BDHS 2022
13a to 13c	Participants and descriptive data	Results (Bivariate Analysis); Appendix D
14a, 14b	Model development	Results (Multilevel Logistic Regression Analysis)
15a, 15b	Model specification	Results (Multilevel Logistic Regression Analysis)
16	Model performance	Results (Internal Validation); External Validation on BDHS 2022
17	Model updating	Not applicable (no updating step)
18	Limitations	Discussion
19a, 19b	Interpretation	Discussion
20	Implications	Discussion; Recommendations
21	Supplementary information	Appendices A to E; Additional files 1 to 3
22	Funding	Declarations

Reference: Moons KGM, Altman DG, Reitsma JB, et al. Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD): explanation and elaboration. *Ann Intern Med.* 2015;162(1):W1-W73.